

UNIT 3 - MICRONUTRIENTS

VITAMINS

Learning objectives

At the end of this unit, students will be able to:

- Define vitamins
- Identify the types of vitamins
- Describe the functions of vitamins
- Explain the characteristics of vitamins
- Understand deficiency syndrome of vitamins

Definition

Vitamins: are defined as organic compounds in food that are needed in very small amounts for growth and for maintaining good health.

They are classified according to the way in which they are discovered and the name implies that they are vital factors in the diet. Many of them cannot be synthesized in the body and must be obtained from the diet.

Classification

Vitamins are classified into two groups;

- Fat soluble vitamins
- Water soluble vitamins

Fat soluble vitamins are vitamins that can only be absorbed in the presence of fat and they include;

- Vitamin A (Retinol)
- Vitamin D (retinol)
- Vitamin E (tocopherol)
- Vitamin K

Water soluble vitamins are quickly absorbed into the body and the excess are excreted in urine. They include the C and B vitamins group.

Initially vitamins were measured in international units (IU), but today they are measured in milligrams (mg). Two groups of compounds that are chemically related to vitamins and are nutritional importance:

- a) **Precursors (Provitamins)** – substances chemically related to vitamins but must be changed by the body into the active forms of vitamins
- b) **Antagonists (Antivitamins)** – also known as pseudovitamins. The body cannot be replaced by real vitamins yet they do not permit reactions. Functionally, several of them are coenzymes, which aid the action of enzymes and also play a role in the many stages in the metabolism of energy yielding nutrients.

Deficiencies could be caused by:

- Lack of nutrients in the diet
- Failure of the body to absorb the nutrient in the diet
- Increased need for a vitamin leading to deficiency symptom

Selecting foods for vitamin content

The vitamin content in food can be reduced due to high temperature, long storage and exposure to air or sunlight. To reduce losses

- The food should be cooked at low temperature, below boiling point and in minimum amounts of water
- The food should be cut into fine pieces with a sharp knife to avoid cell damage
- The cooking water should be added to other dishes.

Vitamin supplements are necessary for healthy persons. Supplementation of the diet may be suitable in cases of emotional illness, which may prevent the consumption of a diet adequate in vitamins

Fat Soluble Vitamins

Fat soluble vitamins A, D, E and K are stored in the body for long periods of time and generally pose a greater risk for toxicity when consumed in excess than water-soluble vitamins. Eating normal, a well-balanced diet will not lead to toxicity in otherwise healthy individuals. However, taking vitamin supplements that contain megadoses of vitamins A, D, E and K may lead to toxicity. The body only needs small amounts of any vitamin. Some health problems may decrease the absorption of fat and in turn, decrease the absorption of Vitamin A, D, E and K.

Characteristics of fat - soluble vitamins

- Metabolize along with fats
- Resistance to heat
- Stored in the liver and adipose tissue
- Slow to develop deficiency syndrome
- Present only in certain foods, mostly in animal products, oily foods, yellow and green vegetable
- Excess can be toxic to the body.

VITAMIN A

It occurs in several forms as:

- a) Retinol
- b) Retinal
- c) Retinoic acid
- d) Retinyl ester

In its pure form, vitamin A is a pure yellow crystalline compound. It occurs naturally in animal and has been synthesized so that it is available commercially. It is soluble in fat and fat solvents but insoluble in water. It is relatively stable to heat, acids and alkalis. It is easily oxidized and rapidly destroyed by UV radiation. It is found in carotenes – yellow pigment synthesized in plants. Man converts diet carotenes to vitamin A.

The carotenes are dark red crystalline compounds. These include Alpha, Beta, Gamma carotene and possibly cryptoxanthin that are of nutritional importance. Upon hydrolysis, each molecule of

the other three yields only one molecule of vitamin A while each molecule of the other 3 yields only one molecule of vitamin A.

Function of vitamin A

1. Maintenance of Normal vision

Vitamin A action in the retina of the eye allows vision in dim light. The retina in blood is oxidized to retinaldehyde or retinal which combines with protein opsin to form rhodospin (pigment visual purple) located in the cells called rods in the retina of the eye and it responsible for the ability of the eye to see in dim light.

2. It aids growth and development during childhood
3. It is essential for normal skeletal and teeth formation. Deficiency will cause bones to grow in length and normal re-modelling process does not take place.
4. Essential for spermatogenesis in men and normal estrous cycle in women. If not available during fetal development many malformations result.
5. Essential for a healthy skin and mucus membranes

Recommended Dietary Allowances

RDA for vitamin A are given as micrograms (mcg) of retinol activity equivalents (RAE) to account for the different bioactivities of retinol and provitamin A carotenoids.

Age	Male	Female	Pregnancy	Lactation
0 – 6 months	400mcg RAE	400mcg RAE		
7 – 12 months	500mcg RAE	500mcg RAE		
1 – 3 years	300mcg RAE	300mcg RAE		
4- 8 years	400mcg RAE	400mcg RAE		
9 – 13 years	600mcg RAE	600mcg RAE		
14 – 18 years	900mcg	700mcg RAE	750 mcg RAE	1,200 mcg RAE
19 – 50 years	900 mcg RAE	700 mcg RAE	770 mcg RAE	1,300 mcg RAE
>50 years	900 mcg RAE	700 mcg RAE		

Food Sources of Vitamin A

Eating a wide variety of foods is the best way to ensure that the body gets enough vitamins A.

Animal source of vitamin A

The best sources of vitamin A is animal products such as organ meat like liver, whole milk and milk products, eggs, butter, cheese, and fish liver oils. Animal sources contain preformed active retinol, which can be used effectively by the body. The best source of vitamin A for infants is breast milk. The mother's secretion of vitamin A into breast milk, however, is related to her own vitamin A status.

Plant source of vitamin A

However, most Africans rely on carotene for their supply of vitamin A. Plants contain beta-carotene that can be converted into vitamin A by the body. The best plant sources of vitamin A are orange, yellow colour fruits and vegetables (papaya, mango, pumpkin, tomatoes, carrots, yellow sweets potatoes) and dark green vegetables such as spinach, kale, amaranthas (michicha), night shades.

Yellow maize is the only cereal that contains carotene. In west Africa, much carotene is obtained from red palm oil, which is widely used for cooking.

Factors Influencing absorption and conversion of carotene

1. Intestine can only absorb and convert a portion of the carotene in the diet. If no animal products are consumed and the body relies entirely on carotene, allowances should be increased.
2. Carotene is poorly utilized on low fat diet and diets deficient in vitamin A are often deficient in fat.
3. Intestinal diseases such as dysentery and colic limit the absorption of vitamin A and the conversion of carotene to vitamin A.
4. Infants and young children do not convert carotene to vitamin A as adults do
5. Bile salts are essential for the absorption of vitamin A and carotene and so persons with obstruction of the bile duct are likely to become deficient in vitamin A.
6. Hydrocarbon oils such as liquid paraffin markedly reduce carotene absorption

Vitamin A Deficiency

Vitamin A deficiency and protein malnutrition are the principle nutritional problems throughout the world. Deficiency of vitamin A can be as a result of:

- Protein deficiency – needed to transport vitamins or to mobilize it from the liver
- Inadequate dietary intake
- Chronic fat malabsorption
- Interference with absorption and storage
- Interference with conversion of carotene to vitamin A

Symptoms show up after the liver reserves have been depleted. The most common signs are:

- Poor growth
- Lowered resistance to infection
- Night blindness
- Rough and scaly skin

What are the consequences of vitamin A deficiency?

Vitamin A deficiency has long been associated with blindness. But more importantly, vitamin A deficiency is associated with increased morbidity and mortality among young children. Improvement of vitamin A levels among deprived populations has been associated with reduction in young child mortality. In developing countries such as Kenya, where diarrhea, acute respiratory infection, and measles are among Nutrition major causes of child mortality, improved vitamin A status is likely to have a much greater role in reducing young child mortality.

The common symptoms of vitamin A deficiency

Night blindness: The child cannot see in the dark. He/she has to go in to the house early in the evening

Conjunctival xerosis: The conjunctival covering the white surface of the eyeballs become dry and rough instead of being moist, smooth and shining. The child cannot open and close his/her eyes because it is painful.

Bitot's spots: A foamy or cheesy accumulation, which forms in the inner quadrant of the cornea in the eyes. The cornea the central transparent part of the eye becomes cloudy. It reflects more advanced vitamin A deficiency, but tends to be reversible with treatment.

Corneal ulceration: an ulcer on the cornea may leave scar, which can affect vision.

Keratomalacia: The eyeballs become opaque and soft, jelly like substance; hereafter there will be a rapid destruction of the eyeball and no hope of recovery after the condition reached the stage of keratomalacia.

Vitamin A toxicity

Vitamin A is not normally excreted and can accumulate in the body. Excessive intake (Hypervitaminosis A) causes:

- Headache
- Dry skin
- Loss of hair
- Softening of bones
- Liver damage

A high incidence of spontaneous abortions and birth defects has been reported in women ingesting large doses of isotretinoin (13-cis-retinoic acid) during the 1st trimester of pregnancy. Carotenoids are not toxic but can colour adipose tissue stores and make the skin look yellow when taken in large doses over several weeks.

VITAMIN D (CHOLECALCIFEROL)

Vitamin D is known as the antirachitic vitamin and chemically as calciferol. The two most important vitamin D compounds are ergocalciferol (vitamin D₂) and cholecalciferol (vitamin D₃). These Nutrition substances are formed from precursors in plants, animal and in the skin and are converted to vitamin D. by the ultraviolet rays of the sun.

Vitamin D is stored in the liver mainly; some is stored in the brain, bones and skin as well. It undergoes changes in the liver, and in the kidneys that convert it to active, hormone like form.

Almost all the vitamin D found naturally in food is vitamin D₃. In foods such as eggs, butter, liver, oily fish and fish liver oil. Irradiation of the plant sterol (ergosterol) e.g. from the yeast yields vitamin D₂, which has similar activity to cholecalciferol in mammals. vitamin D₂ is often used in the production of vitamin supplement foods such as margarine, baby foods and breakfast cereals. Dietary supply of the vitamin becomes much more important in people who do not get adequate exposure to sunlight e.g. household elderly people

Functions of vitamin D

1. Boosts Bone Health. Vitamin D is critical for strong bones from infancy to old age
2. It helps the body absorb calcium from the food. In older adults, a daily source of “D” and calcium helps prevent fractures
3. The presence of vitamin D is essential to the activity of the parathyroid hormone in removing calcium and Phosphorous from the bone in order to maintain normal serum levels of calcium.
4. Stimulates the reabsorption of Calcium by the kidney when serum calcium level is low.
5. Vitamin D plays a role in controlling blood pressure and preventing artery damage
6. Vitamin D may help increase muscle strength, which in turn helps to prevent falls, a common problem that leads to substantial disability and death in older people.

Sources

It can be synthesized in the body with the help of the sunlight.

Food sources

Fortified milk, margarine, butter, juices, cereals and chocolate mixes, egg yolks, liver fatty fish and their oils

Recommended Dietary Allowances

Pediatric

- Infants birth to 12 months: 400 IU (adequate intake)
- Children 1 - 18 years: 600 IU (recommended dietary allowance)

Adult

- 19 - 50 years: 600 IU (recommended dietary allowance)
- 70 years and older: 800 IU (recommended dietary allowance)
- Pregnant and breastfeeding females: 600 IU (recommended dietary allowance)

Deficiency of vitamin D

Leads to inadequate absorption of calcium and phosphorous from the intestinal tract and faulty mineralization of the bone and the teeth structure. Inability of the soft bones to withstand weight stress results into skeletal malformation. In infancy deficiency leads to:

a) Rickets

- Delayed closure of the fontanelle, softening of the skull, bulging forehead giving the head-box like appearance
- Soft fragile bones causing widening of the end bones and bowing of the leg
- Enlargement of the wrist, knee (knock-knees) and ankle joints
- Poorly developed muscles
- Restlessness, nervousness and irritability

b) Tetany

Syndrome is manifested by a sharp flexion of the wrist and ankle, muscle twitching, cramps and convulsions. Results from insufficient dietary calcium or vitamin D, failure to absorb the two or disturbance of the parathyroid gland. Calcium salts are used to control spasms. Liberal calcium diet, vitamin D, concentrates will correct the deficiency

c) Dental carries

Deficiency may lead to delayed dentition and malformation of the teeth and a predisposition to dental carries

d) Osteomalacia

Also called adult rickets. Failure of the process of calcification to keep up with the rest of the metabolic processes caused by vitamin D and calcium. The following changes take place in osteomalacia:

- A softening of the bones which may be so severe that the bones of the legs, spine, thorax, and pelvis bend into deformities
- Pain of rheumatic type in bones of the leg and lower part of the back
- General weakness with difficulties in walking and especially difficulty in climbing stairs
- Spontaneous multiple fractures

Toxicity (Overdose/(Hypervitaminosis D)

Vitamin D is the most toxic of the vitamin and as little as five times the dietary standard (50 microgram/day) may be enough to cause hypercalcaemia when taken for 6 months

Vitamin D poisoning does not occur if the skin is overexposed to the sun because of natural regulation. Poisoning is due to over-fortification of foods or excessive use of supplements.

Excess vitamin D causes hypercalcaemia and this leads to anorexia, nausea vomiting, hypertension, kidney stones, renal failure and calcification of soft tissues

VITAMIN E

Vitamin E activity is exhibited by a number of compounds of related chemical structure including alpha, beta, gamma and delta tocopherol. Alpha- tocopherol is the compound processing most of the vitamin activity biologically.

High temperatures and acids do not affect the stability of vitamin E but oxidation takes place rapidly in the presence of rancid fats, or lead and iron salts. Decomposition occurs in ultra violet light. Vitamin E itself acts on antioxidants. Vitamin E is expressed in International Units (IU) or in mg of alpha-tocopherol.

Functions

1. Vitamin E is an antioxidant that protects body tissue from damage caused by substances called free radicals. Free radicals can harm cells, tissues, and organs. They are believed to play a role in certain conditions related to aging.
2. The body also needs vitamin E to help keep the immune system strong against viruses and bacteria.
3. In the tissues, Vitamin E reduces the oxidation of polyunsaturated fatty acids thus helping to maintain the integrity of the cell membranes
4. It prevents the formation of the lipofuscin in a pigment that forms in the uterus a possible factor in the re-absorption of fetal tissue during miscarriage.
5. Vitamins E is also important in the formation of red blood cells and it helps the body use vitamin K. It also helps widen blood vessels and keep blood from clotting inside them.
6. Synthesis of co-enzyme a factor essential in the respiratory chain that releases energy from carbohydrates or fats.

RDA

Infants (adequate intake of vitamin E)

- 0 to 6 months: 4 mg/day
- 7 to 12 months: 5 mg/day

Children

- 1 to 3 years: 6 mg/day
- 4 to 8 years: 7 mg/day
- 9 to 13 years: 11 mg/day

Adolescents and Adults

- 14 and older: 15 mg/day
- Pregnant teens and women: 15 mg/day
- Breastfeeding teens and women: 19 mg/day

The highest safe level of vitamin E supplements for adults is 1,500 IU/day for natural forms of vitamin E, and 1,000 IU/day for the man-made (synthetic) form.

Food Sources

The best way to get the daily requirement of vitamin E is by eating food sources. Vitamin E is found in the following foods:

- Vegetable oils (such as wheat germ, sunflower, safflower, corn, and soybean oils)
- Nuts (such as almonds, peanuts, and hazelnuts/filberts)
- Seeds (such as sunflower seeds)
- Green leafy vegetables (such as spinach and broccoli)
- Fortified breakfast cereals, fruit juices, margarine, and spreads. Fortified means that vitamins have been added to the food.

Products made from these foods, such as margarine, also contain vitamin E.

Deficiencies

Occur in human beings only when there is a defect in fat absorption as in:

- Cystic fibrosis or in premature infants
- Infants with kwashiorkor
- Increased hemolysis of red blood cells or anaemia

In animals, deficiency has resulted in reproductive failure, macrocytic anaemia (short life span of red blood cells), creatinuria, liver neurosis, encephalomalacia and muscular dystrophy. Vitamin E had been recommended for varying conditions such as heart disease, impotence, ulcers, abortions and menopause.

Toxicity

Too much may cause gastro-intestinal distress following doses of 300 – 600mg daily. In cases of marginal deficiency there may be blood-clotting problem

VITAMIN K

This vitamin can be synthesized by the action of bacteria in the intestinal tract of a healthy person. It is also found in liver, fish, and green vegetables. Cooking does not destroy it. The liver requires vitamin K for the formation of prothrombin a substance needed for clotting mechanism of blood.

Vitamin K consists of a number of related compounds belonging to a chemical group known as quinines.

K1 - Phylloquinone in plant foods

K2 - Menaquinone in animal tissues and synthesized by bacteria

K3 - Menadione synthetic. It is stable to heat and oxygen but light acid, alkali, oxidizing agents and alcohol destroy it

Functions

1. Key function is its use by the liver in the synthesis of various proteins needed for blood clotting. Among these are prothrombin which when its level in blood indicates good ability to coagulate blood and low levels cause slow rate of coagulation
2. Synthesis of osteocalcin a protein found in bone is dependent on vitamin K
3. Important for bone health

Food Sources

The best way to get the daily requirement of vitamin K is by eating food sources. Vitamin K is found in the following foods:

- Green leafy vegetables, such as kale, spinach, turnip greens, collards, Swiss chard, mustard greens, parsley, romaine, and green leaf lettuce
- Vegetables such as Brussels sprouts, broccoli, cauliflower, and cabbage
- Fish, liver, meat, eggs, and cereals (contain smaller amounts)

Vitamin K is also made by the bacteria that line the gastrointestinal tract.

RDA

Infants

0 - 6 months: 2.0 micrograms per day (mcg/day)

7 - 12 months: 2.5 mcg/day

Children

1 - 3 years: 30 mcg/day

4 - 8 years: 55 mcg/day

9 - 13 years: 60 mcg/day

Adolescents and Adults

Males and females age 14 - 18: 75 mcg/day

Males and females age 19 and older: 90 mcg/day

Deficiency of vitamin K

- 1) Low blood levels of prothrombin and other clotting factors leads to increased tendency to hemorrhage. Premature infants and those whose mothers have been taking anti-coagulants are most susceptible to deficiency
- 2) The practice of giving vitamin K3 menadione to mothers prior to delivery resulted in the increase of hemolytic anaemia (breakdown of red blood cells)
- 3) Mental retardation, jaundice, haemorrhaging and neurological symptoms when bile pigment accumulated in the grey matter of the central nervous system
- 4) Deficiency may occur in adults due to failure to absorb or interference of synthesis in the intestine or inability to form prothrombin by the liver

WATER SOLUBLE VITAMINS

They include the ascorbic acid and the B complex vitamins. They are quickly absorbed in the body and the amounts not utilized are excreted in the urine. Since they dissolve in water, large amounts of these vitamins can be found during food processing and preparation.

Characteristics of water soluble vitamins

- They are widely distributed in natural foods
- B12 is found only in animal products
- Soluble in water and absorbed in the intestine
- Excess will be excreted, thus not toxic.
- Most functions of these vitamins are as co-enzymes
- They are important for energy production
- They are heat labile

VITAMIN C

Vitamin C is a water-soluble vitamin that is necessary for normal growth and development.

Water-soluble vitamins dissolve in water. Leftover amounts of the vitamin leave the body through the urine. That means you need a continuous supply of such vitamins in your diet. It is also the most sensitive of all vitamins to heat.

Composition

The activity of vitamin C is possessed by two forms:

1-ascorbic acid (the reduced form)

2- dehydroascorbic acid (oxidized form)

1-dehydroascorbic acid is oxidized further with complete loss of activity. The oxidation is accelerated by heat, light, alkalis, oxidative enzymes, and traces of copper and iron.

Absorption, Metabolism and assimilation

Ascorbic acid is rapidly absorbed from the gastro-intestinal tract and distributed to the various tissues in the body. The adrenal gland and the retina of the eye contain especially high concentration of vitamin C, but other tissues such as the spleen, intestine, bone marrow, pancreas, thymus, liver, pituitary and kidney also contain appreciable amounts. Excessive intake of ascorbic acid does not increase the saturation but is excreted in the urine.

Functions

1. Vitamin C is needed for the growth and repair of tissues in all parts of your body. It is used to:
 - Form an important protein used to make skin, tendons, ligaments, and blood vessels
 - Heal wounds and form scar tissue
 - Repair and maintain cartilage, bones, and teeth
2. Vitamin C is one of many antioxidants. Antioxidants are nutrients that block some of the damage caused by free radicals.
 - Free radicals are made when your body breaks down food or when you are exposed to tobacco smoke or radiation.
 - The buildup of free radicals over time is largely responsible for the aging process.
 - Free radicals may play a role in cancer, heart disease, and conditions like arthritis.
3. Vitamin C enhances iron absorption by keeping iron in its most absorbable form.
4. Vitamin C is vital for the function of the immune system, especially for the activity of certain cells in the immune system
5. Necessary for the synthesis of hormones and other compounds such as bile acids and DNA.
6. Vitamin C is an important partner of protein for tissue synthesis. It is highly concentrated in active tissues in the body indicating its importance in their function e.g. brain, kidney, pancreas, and the spleen

Which foods contain vitamin C?

All fruits and vegetables contain some amount of vitamin C.

Fruits with the highest sources of vitamin C include: Citrus fruits and juices, such as orange and grapefruit, Kiwi fruit, Mango, Papaya, Pineapple, Strawberries, raspberries, blueberries, cranberries, and Watermelon

Vegetables with the highest sources of vitamin C include: Broccoli, Brussels sprouts, cauliflower, Green and red peppers, Spinach, cabbage, turnip greens, and other leafy greens, Sweet and white potatoes, Tomatoes and tomato juice

Some cereals and other foods and beverages are fortified with vitamin C. Fortified means a vitamin or mineral has been added to the food.

Cooking vitamin C-rich foods or storing them for a long period of time can reduce the vitamin C content. Microwaving and steaming vitamin C-rich foods may reduce cooking losses. The best food sources of vitamin C are uncooked or raw fruits and vegetables.

RDA of vitamin C

Adult men (above 19)	90mg/day
Adult women (above 19)	75mg/day
Pregnant women	85mg/day
Lactating mothers	120mg/day

Smokers or those who are around secondhand smoke at any age should increase their daily amount of vitamin C an additional 35 mg per day.

Deficiency

Deficiency of vitamin C results in scurvy. Its first symptoms surface after about a month on a vitamin C free diet. As the body loses its ability to synthesis collagen, connective tissue starts breaking down and gums and joints begin to bleed.

Scurvy occurs in two types; infantile scurvy (in infants) and adult scurvy.

What are the symptoms of these two types of scurvy?

1. Infantile scurvy is an acute scurvy of infants and young children, characterized by periosteal hemorrhage and swelling especially of long bones. Its other symptoms include;

- Pain, tenderness and swelling of the thigh and legs
- Loss of appetite
- Listlessness
- Infant cries when is legs and arms are moved
- Swelling is observed at the ends of long bones
- Hemorrhage occur under the skin
- Gums are swollen and are spongy
- Convulsions may occur, resulting in death of the infants

2. Scurvy in adults

- General weakness
- Spongy and bleeding gums
- Loose teeth
- Swollen tender joints
- Hemorrhage

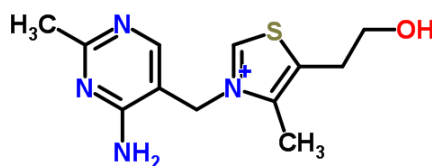
VITAMIN B1 - THIAMINE

It is a water soluble vitamin, also known as the anti-beriberi or anti-neurotic vitamin. The dry vitamin is fairly stable but solutions of it are unstable in the presence of heat and alkali.

Thiamin is also destroyed by oxidation and reduction. It is particularly sensitive to sulphur dioxide, sulphite and so these should not be used to preserve thiamin- containing foods. It is not stored in the body and must be supplied daily.

Structure and composition

Thiamin can exist in foods in a number of forms including free thiamin, the pyrophosphate ester and bound to respective apoenzymes. Thiamin is necessary for catalyzing the oxidation of carbohydrates in the body. This reaction releases energy in the body.



From the structure, it is clear that thiamine contains nitrogen, hydrogen, carbon, oxygen and sulphur atoms in it.

Food Sources

Nearly all foods, except sugars, fats and oils contain some thiamine. The best sources are cereal grain (particularly wheat) germ layers.

Other sources include; pulses; peas and beans, nuts, liver, lean meats, poultry, egg yolk, fish, dried yeast and milk.

FUNCTIONS

1. The most important function of thiamine is its role as a coenzyme. It combines with phosphoric acid to form thiamine pyrophosphate (TPP) which functions as a coenzyme. TPP is the coenzyme of carboxylase which is required for oxidative decarboxylation of pyruvate during the biological oxidation of glucose.
2. Thiamin (vitamin B1) helps the body's cells change carbohydrates into energy. The main role of carbohydrates is to provide energy for the body, especially the brain and nervous system
3. Thiamin also plays a role in muscle contraction and conduction of nerve signals.
4. It is essential to maintain the nerve in healthy condition.
5. It is essential for the maintenance of good appetite and normal digestion.
6. It is required for the normal growth of children and for general health.

Absorption, metabolism and assimilation

Thiamin is absorbed in small intestines. It is released by the action of phosphatase and pyrophosphatase in the upper small intestines. Active transport is greatest in jejunum and ileum (it is inhibited by alcohol consumption and folic deficiency). Decline in thiamin deficiency occurs at intakes above 5mg. thiamine, in its active form, (thiamine pyrophosphate, TPP), serves as a coenzyme in a variety of reactions, particularly in carbohydrate metabolism. It plays a key role in the metabolic breakdown of glucose to yield energy in body tissues. About half of the body stores of thiamin are found in skeletal muscles, with the remainder dispersed throughout the heart, liver, kidneys and nervous tissues, including the brain.

RDA of thiamine

The RDA of thiamine varies according to the intake of calories, being approximately 0.5mg per 1000calories. The average adult male needs approximately 1.2mg/day, while the adult female needs about 1.1mg/day.

Deficiency

A deficiency of thiamine affects chiefly the nervous system, the circulation, and the gastro intestinal tract. Symptoms include;

- Loss of appetite
- **Mental** confusion
- Muscular weakness
- Pain in the calf muscles
- Emotional disturbances
- Irritability
- Increased pulse rate
- Dyspnea
- Reduced intestinal mobility
- Heart irregularities
- Multiple neuritis

Prolonged deficiencies lead to beriberi, a serious disease involving damage to the nervous system and the heart. There are three forms of beriberi namely;

- i. Wet beriberi
- ii. Dry beriberi
- iii. Infantile beriberi

Now, let us look at each of them individually.

Dry beriberi

The signs and symptoms of this disease include the following;

- Loss of appetite
- Tingling and numbness of the legs and hands
- Wasting of muscle and difficulty in walking

Wet beriberi

- Symptoms of dry beriberi(name them)
- Oedema in the legs
- Enlargement of the heart
- Palpitation and breathlessness.

Infantile beriberi

Early symptoms include;

- Restlessness
- Sleeplessness
- Loss of appetite

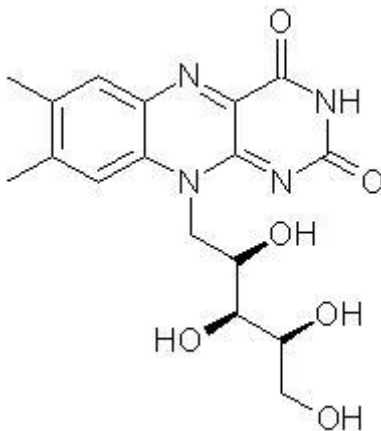
- Palpitation, breathlessness and loss of appetite develop, as the disease advances, due to the enlargement of the heart. Death may occur suddenly if treatment is delayed.

Wernicke's encephalopathy (seen often in alcoholics) is characterized by ophthalmoplegia, polyneuritis, ataxia and mental deterioration. It occurs occasionally in people who fast. In thiamine deficiency, there is accumulation of pyruvic and lactic acid in the tissues and body fluids.

Fortunately, thiamine deficiency is not very common, although it may be rather prevalent among the homeless, alcoholics and other special groups. Thiamine deficiency occurs as a result of many factors including crash dieting, alcohol abuse, liver dysfunction kidney dialysis, and sustained periods of IV (intravenous) nutrients. Also at risk are those who consume a lot of sweets, soft drinks, and highly processed foods.

VITAMIN B₂ RIBOFLAVIN

It is a water soluble yellow pigment. It is less water soluble than thiamine and more stable to heat. When in solution, the vitamin is destroyed on exposure to sunlight. The vitamin is composed of hydrogen, carbon, oxygen, nitrogen and atoms.



What are some of the characteristics of Riboflavin?

- It is yellow in color
- On reduction, it changes into a colorless form
- It is less water soluble than thiamin
- It is destroyed by heating in alkaline solution

- Owing to heat stability and limited water solubility, very little of the vitamin is lost during the cooking and processing of food.
- Baking soda used for faster cooking destroys much of the riboflavin content of the food.
- On exposure to light, riboflavin readily loses its vitamin activity due to the photochemical cleavage of the ribitol moiety of the vitamin, forming lumiflavin.

Dietary Sources

The best sources of riboflavin include brewer's yeast, almonds, organ meats, whole grains, wheat germ, wild rice, mushrooms, soybeans, milk, yogurt, eggs, broccoli, Brussels sprouts, and spinach. Flours and cereals are often fortified with riboflavin.

Riboflavin is destroyed by light, so food should be stored away from light to protect its riboflavin content. While riboflavin is not destroyed by heat, it can be lost in water when foods are boiled or soaked. During cooking, roasting, and steaming preserves more riboflavin than frying or scalding.

Why is vitamin B2 necessary?

1. Riboflavin is involved in vital metabolic processes in the body, and is necessary for energy production and normal cell function and growth.
2. It is also crucial in helping other B vitamins undergo the chemical changes that make them useful.
3. Emerging research shows that riboflavin can act as an antioxidant, potentially helping to prevent cancer and prohibit cholesterol buildup by controlling the proliferation of harmful molecules known as free radicals.
4. Common diseases and illnesses that may be prevented or alleviated via riboflavin supplementation include neonatal jaundice, anemia, anorexia/bulimia, cataracts, cognitive function (thinking and memory), depression, and migraines.

What are the signs of a deficiency?

Riboflavin deficiency (called ariboflavinosis) can appear at intakes of less than 0.5-0.6 mg/day. Too little riboflavin can cause weakness, throat swelling/soreness, a swollen tongue, skin cracking (including cracked corners of the mouth), dermatitis, and anemia. Riboflavin deficiency

can also affect vision, including blurred vision and itching, watering, sore, or bloodshot eyes, as well eyes becoming light-sensitive and easily fatigued.

Particular groups may be especially susceptible to riboflavin deficiency, including the elderly, the chronically ill and alcoholics. Women who take birth control pills may also benefit from supplementation - the body's ability to absorb riboflavin is reduced when taking birth control pills.

How much, and what kind, does an adult need?

The Recommended Dietary Allowance (RDA) for is 1.0 mg for female adolescents between the ages of 14-18 years; 1.3 mg for male adolescents 14-18 years of age; 1.1 mg for female adults older than 18 years; and 1.3 mg for male adults older than 18 years. Pregnant women are recommended to take 1.4 mg, and women who are breast feeding should take 1.6 mg. People who are at a higher risk for riboflavin deficiency (see above) should talk with a physician for adequate dosages.

How much does a child need?

The NIH says the RDA of riboflavin for infants and children is 0.3 mg for 0-6 months old; 0.4 mg for 7-12 months old; 0.5 mg for 1-3 years old; 0.6 mg for 4-8 years old; and 0.9 mg for 9-13 years old.

How do you get enough vitamin B2 from foods?

Healthy individuals who eat a balanced diet may not need to supplement with riboflavin. Dietary sources of riboflavin include: dairy products (such as milk, cheese and yogurt), eggs, enriched or fortified cereals and grains, meats, liver, dark greens (such as asparagus, broccoli, spinach and turnip greens), fish, poultry, and buckwheat. Keep in mind that riboflavin is easily destroyed by exposure to light, so buy milk and yogurt in paper cartons or containers.

Are there any risks associated with too much vitamin B2?

Excess riboflavin is excreted through the urine (often temporarily causing urine to turn a bright yellow color several hours after taking it), so there is no common risk for an overdose. However, extremely high doses may result in an increased risk of kidney stones. Sensitivity to light, itching, numbness, and burning/prickling sensations may also occur at high dosages. Allergy and anaphylaxis (severe, whole-body allergic reaction) have also been reported, though rarely.

VITAMIN B₃ - NIACIN

Vitamin B₃, also known as niacin, is actually made up of niacin (nicotinic acid) and niacinamide. It is a water-soluble nutrient that is part of the B vitamin family. B vitamins help to support adrenal function, to calm and maintain a healthy nervous system, and are necessary for key metabolic processes. Niacin is essential for the metabolism of carbohydrates and fats.

Why is vitamin B₃ necessary?

1. Niacin's reduces the risk of heart disease and lowers harmful cholesterol while raising good cholesterol.
2. Niacin may also reduce the incidence of asthma-induced wheezing, and may be helpful in treating or preventing atherosclerosis, second heart attacks, Alzheimer's disease and osteoarthritis.
3. Niacin also helps the body make various sex and stress-related hormones in the adrenal glands and other parts of the body
4. Niacin helps improve circulation.

What are the signs of a deficiency?

Pellagra is a nutritional disease that develops due to insufficient dietary amounts of vitamin B₃. Symptoms of pellagra include skin disease, diarrhea, dementia and depression.

How much, and what kind, does an adult need?

The Recommended Daily Allowance (RDA) for adults is between 16 and 18 mg daily, with a maximum intake of 35 mg daily. It is recommended that pregnant or breastfeeding women avoid supplemental niacin.

How do you get enough Vitamin B₃ from foods?

The best food sources of vitamin B₃ are found in beets, brewer's yeast, beef liver, beef kidney, fish, salmon, swordfish, tuna, sunflower seeds, peanuts, Salmon and tuna, eggs, leafy vegetables, broccoli, tomatoes, carrots, sweet potatoes, avocados, nuts, whole grains, legumes and mushrooms

Bread and cereals are usually fortified with niacin. In addition, foods that contain tryptophan, an amino acid the body converts into niacin, include poultry, red meat, eggs, and dairy products

Are there any risks associated with too much vitamin B₃?

It is difficult to get too much niacin from food sources, but you can get too much niacin when supplementing. Reactions range from flushing, itching, nervousness and headaches to intestinal cramps, nausea, and diarrhea. High doses of niacin can cause liver toxicity; doses in excess of 3 grams a day should be used only under careful medical supervision (for example, in the treatment of high cholesterol). Gout, abnormal heart rhythms and worsening of stomach ulcers have also been reported with very high doses of supplemental vitamin B₃.

VITAMIN B5 - PANTOTHENIC ACID

Vitamin B5, more commonly known as pantothenic acid, is a water-soluble nutrient that is part of the B vitamin family. The B vitamins help support adrenal function and a healthy nervous system. The vitamin's name derives from the Greek word *pantotheron*, meaning "from everywhere," reflecting the fact that small amounts of pantothenic acid can be found in nearly every food.

Why is it necessary?

1. Pantothenic acid is vital in the metabolism of proteins, carbohydrates, and fats;
2. Vitamin B5 is critical to the manufacture of red blood cells, as well as sex and stress-related hormones produced in the adrenal glands, small glands that sit atop the kidneys.
3. Vitamin B5 is also important in maintaining a healthy digestive tract, and it helps the body use other vitamins, particularly B2 or riboflavin.
4. Is used in the synthesis of coenzyme A (CoA), an enzyme that participates in a variety of reactions in the body, especially the breaking down of fatty acids.

What are the signs of a deficiency?

Pantothenic acid deficiency is a rare but serious condition that can cause numbness and burning sensations in the hands and feet as well as headaches, fatigue, and insomnia. Since pantothenic acid is involved in a wide variety of biological functions, deficiencies of the vitamin may not be easily identified or may be masked by other nutrient deficiencies.

How much, and what kind, does an adult need?

A daily adequate intake (AI) of 5 mg for both men and women is recommended. The recommended daily AI for pregnant women is 6 mg, and seven mg for breastfeeding women. There is no known toxicity level for pantothenic acid in humans, but gastrointestinal side effects such as nausea and heartburn have been reported in doses up to 1,200 mg.

N//B: When data are insufficient to set an Estimated Average Requirement (EAR) for a nutrient, an RDA cannot be established. In these cases, an Adequate Intake (AI) level is set.

How do you get enough from foods?

Pantothenic acid is a common nutrient that is found in a variety of foods in at least a small amount. The highest amounts of pantothenic acid are found in whole grain cereals, eggs, meat, legumes, avocado, and yogurt. Studies have shown processing and refining grains result in large losses of pantothenic acid, in some cases up to 75 percent, but since the nutrient is found in a wide array of foods, there is little risk of deficiency. The only documented cases have been in conjunction with severe malnutrition.

Are there any risks associated with consuming too much?

The only known adverse effect is diarrhea as a result of very high intakes - between 10 and 20 grams - of calcium D-pantothenate, a common supplement. Gastrointestinal side effects, such as heartburn and nausea, have also been noted with pantothenic acid doses up to 1,200 mg.

VITAMIN B6 – PYRIDOXINE

Vitamin B6, also called pyridoxine, is a water-soluble nutrient that is part of the B vitamin family. B vitamins help support adrenal function, help calm and maintain a healthy nervous system, and are necessary for key metabolic processes. Vitamin B6 acts as a coenzyme in the breakdown and utilization of carbohydrates, fats and proteins.

Why is vitamin B6 necessary?

1. Vitamin B6 helps in the production of neurotransmitters, the chemicals that allow brain and nerve cells to communicate with one another, ensuring that metabolic processes such as fat and protein metabolism run smoothly, and is important for immune system function in older individuals.

2. It is needed for normal brain development and function, and helps the body make the hormones serotonin and norepinephrine, which influence mood, and melatonin, which helps regulate the body clock.
3. Along with vitamins B12 and B9 (folic acid), B6 helps control levels of homocysteine in the blood. Homocysteine is an amino acid that may be associated with heart disease.
4. Your body needs B6 in order to absorb vitamin B12 and to make red blood cells and cells of the immune system.

What are the signs of a deficiency?

Vitamin B6 deficiency can lead to nerve damage in the hands and feet. Cervical dysplasia has been linked to a low intake of several B vitamins including pyridoxine, and people with alcoholism, cirrhosis, hyperthyroidism and congestive heart failure may experience deficiencies more often. Some symptoms of a vitamin B6 deficiency include dermatitis, cracked and sore lips, inflamed tongue and mouth, confusion, depression and insomnia.

How much, and what kind, does an adult need?

Recommended Daily Allowance (RDA) for adult males between 19 and 50 years of age is 1.3 mg, and those over the age of 50 need 1.7 mg. Women between 19 and 50 years of age should take 1.3 mg, and those over 50 should take 1.5 mg. Pregnant women should take 1.9 mg and lactating women, 2 mg.

How much does a child need?

infants get 0.1 mg per day, children from 7 to 12 months get 0.3 mg; children between 1 and 3 years of age get 0.5 mg. Children from 4 to 8 years old should get 0.6 mg; from 9-13 years, 1 mg; teenage males 14-18 years old 1.0 mg per day, teenage females 4-18 years old 1.2 mg per day.

How do you get enough vitamin B6 from foods?

Good food sources of vitamin B6 include brewer's yeast, bananas, cereal grains, legumes, vegetables (especially carrots, spinach and peas), potatoes, milk, cheese, eggs, fish and sunflower seeds.

Are there any risks associated with too much vitamin B6?

The current recommended maximum daily intake is 100 mg. High doses of vitamin B6 can, over time, be toxic, and may result in nerve damage or numbness and tingling in the extremities that may eventually be irreversible. You should discontinue use of supplemental B6 if any unusual numbness develops in the body. Too much B6 can also cause oversensitivity to sunlight, which can lead to skin rashes and numbness, as well as nausea, vomiting, abdominal pain, loss of appetite, and increased liver function test results.

VITAMIN B 9 – FOLATE

Vitamin B9, more commonly known as folate or folic acid, is a water-soluble vitamin that is part of the B vitamin family. B vitamins help support adrenal function, help calm and maintain a healthy nervous system, and are necessary for key metabolic processes. Folate occurs naturally in foods, while folic acid is the synthetic form of folate.

Why is vitamin B9 necessary?

1. Vitamin B9 is essential for human growth and development, encourages normal nerve and proper brain functioning,
2. Folic acid is crucial for proper brain function and plays an important role in mental and emotional health.
3. It aids in the production of DNA and RNA, the body's genetic material, and is especially important when cells and tissues are growing rapidly, such as in infancy, adolescence, and pregnancy.
4. Folic acid also works closely with vitamin B12 to help make red blood cells and help iron work properly in the body.
5. Vitamin B9 works with vitamins B6 and B12 and other nutrients to control blood levels of the amino acid homocysteine. High levels of homocysteine are associated with heart disease.

Pregnant women have an increased need for folic acid: it supports the growth of the placenta and fetus, and helps to prevent several types of birth defects, especially those of the brain and spine. Pregnant women and women of child-bearing age should take extra caution to get enough folic acid

What are the signs of a deficiency?

Deficiency has been linked to birth defects, low birth weight, pregnancy loss, depression, memory loss, and cervical dysplasia. Alcoholics, pregnant women, and people living in institutional settings are at a higher risk of vitamin B9 deficiency. Folate deficiency may result to macrocytic or megaloblastic anaemia.

How much, and what kind, does an adult need?

Recommended Dietary Allowance (RDA) is 400 micrograms for adults. During pregnancy the allowance is 800mcg and for lactation it is 600mcg. During the 1st year of life the needs are met with 50mcg daily.

How do you get enough vitamin B9 from foods?

Spinach, green vegetables and beans are good sources, as are fortified products such as orange juice, baked goods, and cereals. Other natural sources of folate include asparagus, bananas, melons, lemons, legumes, yeast, and mushrooms.

Are there any risks associated with too much vitamin B9?

Folic acid has few side effects, even when taken in high amounts. Although the folic acid itself is not a problem, supplemental folic acid can mask symptoms of pernicious anemia, a potentially fatal disease which is caused by a deficiency of vitamin B12. Very high doses (above 15,000 mcg) can cause stomach problems, sleep disturbances, skin reactions, and seizures.

VITAMIN B12 - CYANOCOBALAMIN

Vitamin B12, also known as cyanocobalamin, is a water-soluble vitamin that is part of the B vitamin family. B vitamins help support adrenal function, help calm and maintain a healthy nervous system, and are necessary for key metabolic processes. Vitamin B12 is important to DNA synthesis and maintaining healthy nerve cells.

Why is vitamin B12 necessary?

1. Vitamin B12 is an especially important vitamin for maintaining healthy nerve cells, and it helps in the production of DNA and RNA, the body's genetic material.
2. Vitamin B12 also works closely with vitamin B9, also called folate or folic acid, to help make red blood cells and to help iron work better in the body.
3. Folate and B12 work together to produce S-adenosylmethionine (SAME), a compound involved in immune function and mood.
4. Vitamins B12, B6, and B9 work together to control blood levels of the amino acid homocysteine. High levels of homocysteine are associated with heart disease. However, researchers aren't sure whether homocysteine is a cause of heart disease or just a marker that indicates someone may have heart disease.

What are the signs of a deficiency?

Symptoms of vitamin B12 deficiency include fatigue, muscle weakness, shortness of breath, dizziness, numbness, heart palpitations, bleeding gums and mouth sores, nausea, poor appetite and diarrhea. Symptoms may present themselves slowly and may not be recognized for some time. A deficiency of B12 can produce pernicious anemia, which can lead to memory loss, confusion and even dementia.

Since we obtain vitamin B12 only from animal foods in our diet, deficiencies tend to develop among strict vegetarians, especially vegan children, who eat no animal products. However, the elderly, and those who are unable to absorb vitamin B12 from the intestinal tract are also at risk, as well as those who are pregnant or who suffer hemorrhage or intestinal disorders.

How much, and what kind, does an adult need?

Recommended Dietary Allowance (RDA) is:

- Age 14 and older, 2.4 mcg
- Adult and adolescent pregnant females, 2.6 mcg
- Adult and adolescent lactating females, 2.8mcg

People over 50 years of age should consume vitamin B12-fortified foods, or take a vitamin B12 supplement - 25-100 mcg per day has been used to maintain vitamin B12 levels in older people.

How do you get enough vitamin B12 from foods?

Vitamin B12 is found only in animal foods. Good dietary sources include fish, shellfish, dairy products, organ meats -- particularly liver and kidney-- eggs, beef, and pork.

Are there any risks associated with too much vitamin B12?

Vitamin B12 is considered safe and non-toxic, but other B-vitamins can cause problems in large doses.

Discussion questions

1. Define vitamins.
2. State the general functions of vitamins.
3. What are the characteristics of water and fat soluble vitamins?
4. Why vitamin A deficiency disease is a very serious health problem?
5. How can we control and prevent vitamin deficiency diseases?

UNIT THREE

MINERALS

Learning objectives:

At the end of this unit, students will be able to:

- Define minerals
- Describe the functions of minerals
- Understand the sources of minerals
- Identify the deficiency of minerals

Minerals are inorganic substances present in all body tissues and fluids. They occur in foods as salts and also in combination with organic substances. Minerals have two distinct characteristics

- i) They do not provide energy
- ii) They are not destroyed during food preparation

Mineral elements in the body form 4-6% of the total weight of an adult. Minerals are categorized based on the amount we need per day. If more than 100 milligrams of a mineral are required per day then it is considered a major mineral

General Functions of Minerals

1. Bone and teeth formation – the largest concentration of minerals is found in the bones and teeth
2. Muscle contraction and response of nerves e.g. calcium
3. Control of water balance e.g. Na and K
4. Maintenance of Acid-Base balance
5. Components and co-factors of vitamins, hormone and enzymes e.g. sulphur is part of thiamin; B12 contains cobalt
6. Clotting of blood

Factors that favour mineral absorption

a) Body requirements

If the body needs are high e.g. in pregnancy and in growth stage of children, there is higher absorption of calcium and iron from the diet.

b) Stomach acidity

Calcium and iron absorption is improved in an acidic condition. Absorption of calcium in the intestinal tract is enhanced by lactose which is hydrolysed to lactic acid. Ascorbic acid improves the absorption of calcium, iron and zinc.

c) Bioavailability

This is the degree to which the amount of ingested nutrient is absorbed and is available to the body. The form in which the mineral is present in haeme iron in animal foods is more easily absorbed in the body than iron compounds present in plant foods.

Factors that Hinder absorption of minerals elements in the body

1. Fiber – Mineral interaction

Components of fibre especially phytic acid in grain bind minerals to form insoluble complexes hence reducing absorption. If grains are leavened with yeast enzymes produced helps to break the bonds hence increased mineral absorption. Fibre diets reduce absorption of zinc, iron and magnesium.

2. Mineral – Mineral interaction

Absorption is adversely affected by an excessive intake of other minerals. Many minerals are of similar sizes and chemical charges which causes minerals to compete with each other for absorption hence affecting each other bioavailability e.g. Magnesium, calcium, iron and copper. Intake of iron supplements hinders absorption of zinc. Zinc supplements reduce the absorption of copper

3. Vitamin – Mineral interaction

Vitamin C promotes iron absorption in the body. Active vitamin D hormone improves calcium absorption. Many vitamins require specific minerals to act as components in their structure and function e.g. thiamin co-enzyme requires magnesium and manganese to function efficiently

4. Intestinal mobility

Increased mobility of intestinal track due to diarrhea, intake of laxatives etc reduce the time of absorption and hence the amount absorbed

5. Parasites

Parasites present in the intestinal tract also interfere with the absorption of minerals. Iron deficiency anaemia in children is attributed to this factor.

Toxicity

Toxicity due to excessive intake of minerals from normal natural foods is very remote. It possibly due to:

- Iron or other nutritional supplements taken in excess
- Errors made in substituting supplements
- Accidental consumption e.g. by children
- Exposure to toxic levels of chemicals in an industrial plant

MACROMINERALS

CALCIUM

Calcium is the most abundant mineral in the body, present mainly in the bones and teeth. It is an essential dietary element required in optimal amounts for good bone health, efficient nerve and muscle function, and overall cardiovascular health. It makes up about three percent of the earth's crust and is a basic component of most animals and plants. Our bones serve as a storage site for the body's calcium, providing this mineral to the bloodstream for use by the heart and other organs. Eating a diet rich in calcium helps to restore it to the bones.

Why is calcium important to our bodies?

1. Forming and maintaining bones and teeth are calcium major role in the body
2. It is essential for the blood clotting process
3. It is involved in normal muscle contraction which includes heartbeat
4. Control of the transmission of nerve impulses
5. Influences the activities of various enzymes and hormonal responses e.g. pancreatic lipase as a co-factor
6. Ensure the absorption of vitamin B12
7. Maintenance of permeability of cell membranes to permit movement of material in and out of the cells. A drop in normal concentration of blood calcium produces cramp in the muscles a characteristics muscle twitching is experienced

What are the signs of a deficiency?

Symptoms of calcium deficiency (also known as hypocalcemia) range from minor - numbness or tingling of the fingers, muscle cramps, lethargy and poor appetite - to more severe, including mental confusion, skeletal malformations, dermatitis, and in infants, delayed development.

Illnesses such as osteoporosis (brittle, thin, porous bones that easily break) and rickets are also associated with a deficiency.

How much, and what kind, does one need?

Children	525-550mg/day
Males	700-1000mg/day
Females	800-700mg/day
Pregnant	700mg/day
Lactating	1250mg/day

How do you get enough from foods?

The richest food sources of calcium include cheeses, dairy products, such as milk and yogurt; tofu; and blackstrap molasses. Some other good sources of calcium include almonds, brewer's yeast, Brazil nuts, broccoli, cabbage, dried figs, kelp, dark leafy greens such as kale, hazelnuts, oysters, sardines, and canned salmon. Foods that are fortified with calcium, such as juices, soy milk, rice milk, tofu and cereals, are also good sources of this mineral.

Are there any risks associated with too much?

Calcium supplements can be constipating, and should be balanced with magnesium as discussed above. Excessive amounts in the blood may have negative effects, including nausea, vomiting, loss of appetite, and increased urination. More serious complications include kidney toxicity, confusion, and irregular heart rhythm. Studies indicate that men who take too much may have an increased risk of prostate cancer, and should limit their dietary intake to 500-600 mg daily from all sources.

SODIUM

Sodium is a chemical element that makes up 40 percent of table salt (or sodium chloride) by weight. Salt is the number-one source of sodium in our diet.

Why is sodium necessary?

1. Sodium plays a critical role in our bodies by helping regulate water balance
2. Sodium is needed for the muscles and nerves to work properly.
3. Keeping the circulatory system functioning properly. Our bodies require sodium to regulate blood pressure and protect against blood clotting.
4. It also plays a role in the digestive system, helping to metabolize food.
5. Regulation of acid-base balance of the body
6. Regulations of the osmotic pressure of the plasma or tissue fluids

What are the signs of a deficiency?

Sodium deficiency is rare. Only those who engage in vigorous exercise in hot environments need be concerned about taking in extra sodium to compensate for the amount lost through perspiration.

How much, and what kind, does an adult need?

Adults need only about 1,500-2,000 mg of sodium per day, although some research indicates 1,000 mg may be sufficient.

How much does a child need?

Children need only between 1,000 and 1,300 mg a day.

How do you get enough sodium from foods?

Sodium occurs naturally in most foods. The most common form of sodium is sodium chloride, which is table salt. Milk, beets, and celery also naturally contain sodium, as does drinking water, although the amount varies depending on the source.

Sodium is also added to various food products. Some of these added forms are monosodium glutamate, sodium nitrite, sodium saccharin, baking soda (sodium bicarbonate), and sodium

benzoate. These found in items such as Worcestershire sauce, soy sauce, onion salt, garlic salt, and bouillon cubes.

Processed meats, such as bacon, sausage, and ham, and canned soups and vegetables are all examples of foods that contain added sodium. Fast foods are generally very high in sodium.

Are there any risks associated with too much sodium?

Excess dietary sodium has been linked to a variety of health issues, ranging from fluid retention and circulatory problems to high blood pressure, heart, or kidney disease. People at risk for these problems may benefit from a low-sodium diet.

Potassium and sodium balance each other - it is the ratio of sodium to potassium in the diet and in our systems that seems to affect blood pressure and kidney function more than sodium levels alone.

POTASSIUM

Potassium is an essential mineral. Like sodium, it binds readily with other minerals, and does not occur naturally in an unbound state. Potassium is required for the proper functioning of many major organ systems.

Why is potassium necessary?

1. Potassium is essential for the heart, kidneys, muscles, nerves, and digestive system to operate normally
2. For regulating fluid balance,
3. Regulating the body's acid-base balance, and blood pressure.
4. Regulation of PH of cell contents
5. Regulation of the osmotic pressure of cell contents

What are the signs of a deficiency?

Potassium deficiency symptoms include irregular heartbeat, muscle weakness and mood changes, as well as nausea and vomiting. The resulting condition is known as hypocalcemia. People with kidney disease, gastrointestinal disease and those who take diuretics may have lower levels of potassium.

How much, and what kind, does an adult need?

Adults age 19 and older: 4.7 g/day. Women who are producing breast milk need slightly higher amounts (5.1 g/day).

How do you get enough potassium from foods?

Many foods contain potassium. All meats (red meat and chicken) and fish such as salmon, cod, flounder, and sardines are good sources of potassium. Soy products and veggie burgers are also good sources of potassium. Vegetables including broccoli, peas, beans, tomatoes, potatoes (especially their skins), sweet potatoes, are all good sources of potassium.

Fruits that contain significant sources of potassium include citrus fruits, bananas, kiwi, prunes, and apricots. Dried apricots contain more potassium than fresh apricots. Milk and yogurts, as well as nuts, are also excellent sources of potassium.

People with kidney problems, especially those on dialysis, should not eat too many potassium-rich foods. The doctor or nurse will recommend a special diet.

Are there any risks associated with too much potassium?

Too much potassium (hyperkalemia) can cause upset stomach, vomiting and diarrhea. More serious side effects include irregular heartbeat, mental confusion, tingling or burning sensations in the extremities, and gray skin.

PHOSPHORUS

Phosphorus is a mineral. Highly reactive, it is never found in its free form in nature. Present as phosphate in every cell of the body, it is a nutrient required by all organisms for the basic processes of life, from energy storage to the formation of bones and teeth.

Why is phosphorus necessary?

1. Healthy bones and soft tissues require phosphorus (along with calcium) to grow and develop throughout life.
2. Optimum phosphate levels also promote healthy metabolism, the utilization of many B-complex vitamins, proper muscle and nerve function, and calcium balance.
3. Phosphates are used to treat a variety of ailments, including constipation, kidney stones, and high blood calcium levels.

4. Essential for the storage and controlled release of energy from ADP – ATP (Adenosine Triphosphate)
5. Inorganic phosphates in the body fluids constitute an important buffer system in the regulation of the body neutrality. The level is kept constant by the kidney

What are the signs of a deficiency?

Muscle weakness is the most common symptom of phosphate deficiency; this can include impairment of heart contraction. Other symptoms may include confusion, seizures and coma.

How much, and what kind, does an adult need?

Adults aged 18 years and older, including pregnant or breastfeeding women get 700 milligrams of phosphorus per day. Infants 0-6 months-old of 100 mg per day (additional phosphorus may be added to infant formulas); for infants 7-12 months old, 275 milligrams per day; for children 1-3 years old, 460 milligrams per day; children between 4 and 8 is 500 milligrams per day; and children ages 9-18 years old is 1,250 milligrams per day.

How do you get enough phosphorus from foods?

The main food sources are the protein food groups of meat and milk. A meal plan that provides adequate amounts of calcium and protein also provides an adequate amount of phosphorus. Although whole-grain breads and cereals contain more phosphorus than cereals and breads made from refined flour, this is a storage form of phosphorus called phytin, which is not absorbed by humans. Fruits and vegetables contain only small amounts of phosphorus.

Are there any risks associated with too much phosphorus?

Excess phosphate intake can result in hyperphosphatemia (high blood phosphorus levels) which can lead to potentially serious electrolyte imbalances and even death.

Excessive intake of phosphorus (or phosphate) supplements may worsen conditions such as heart disease, pancreatitis, rickets, osteomalacia (softening of bones), high blood pressure, and liver and kidney disease. Talk with your physician before taking phosphorus supplements if you have any health conditions.

MAGNESIUM

Magnesium is needed for more than 300 biochemical reactions in the body. It helps to maintain normal nerve and muscle function, supports a healthy immune system, keeps the heart beat steady, and helps bones remain strong.

Functions

1. It also helps regulate blood glucose levels and aid in the production of energy and protein.
2. It also contributes to the makeup of teeth and bones.
3. It activates enzymes, contributes to energy production
4. Helps regulate calcium levels, as well as copper, zinc, potassium, vitamin D, and other important nutrients in the body.

Absorption, metabolism, assimilation and excretion

The average intake from the diet by adults is about 300-400mg. a greater part of this (40-50%) is not absorbed and hence excreted in the stools.

About one-third of the amount ingested is excreted in the urine. The magnesium content of normal human serum is about 2 to 3mg/100ml and of whole blood 1.56mg per 100ml.

RDA

Adults – 200 – 300mg/day

Older children 150 – 200mg/day

Infants and preschool children 100-150mg/day

Most dietary magnesium comes from vegetables, such as dark green, leafy vegetables. Other foods that are good sources of magnesium:

- Fruits or vegetables (such as bananas, dried apricots, and avocados)
- Nuts (such as almonds and cashews)
- Peas and beans (legumes), seeds
- Soy products (such as soy flour and tofu)
- Whole grains (such as brown rice and millet)

Deficiency

Symptoms occur when magnesium level is below 1mg/100ml include: depression, muscular weakness and liability to convulsions. Magnesium deficiency has been observed in chronic alcoholics as indicated by low serum mg and muscular weakness. In children, it is possible that

magnesium deficiency exists in cases of kwashiorkor as the serum magnesium level has been reported to be low.

TRACE MINERALS/MICROMINERALS

IRON

Iron is an essential mineral that is required for human life. Much of the iron in the body is found in red blood cells and carries oxygen to every cell in the body. Iron also is involved in producing ATP (adenosine triphosphate, the body's energy source). Extra iron is stored in the liver, bone marrow, spleen, and muscles.

Functions of Iron

1. Formation of haemoglobin and blood
2. Brain development and function
3. Sound immune system. Lack of iron diminishes the number of T-cells and the production of antibodies.
4. Iron is a component of myoglobin, catalase and certain enzyme systems
5. Carrier of oxygen and carbon dioxide as part of haemoglobin and myoglobin and part of several tissue enzymes
6. Anti-effective agent. Lactoferrin and iron containing compound present in human milk that binds iron making it unavailable for E-coli organisms in the G.I tract of the infant

Dietary Sources

There are 2 types of iron you can get from food: heme and non heme. Heme iron is more easily absorbed by the body. The best dietary sources of heme iron are liver and other organ meats, lean red meat, poultry, fish, and shellfish (particularly oysters).

Sources of non heme iron include dried beans and peas, legumes, nuts and seeds, whole grains, dark molasses, and green leafy vegetables. Some nutrients help the body better absorb this kind of iron. For example, vitamin C helps the absorption of this type of iron while calcium (including all dairy products), bran, tea, and unprocessed whole grain products block its absorption.

RDA

Adult

- Male 19 - 50 years: 8 mg daily
- Female adults 19 - 50 years: 18 mg daily
- Adults 51 years and older: 8 mg daily
- Pregnant females ages 14 - 50 years: 27 mg daily
- Nursing females ages 14 - 18 years: 10 mg daily
- Nursing females ages 19 - 50 years: 9 mg daily

Effective of Imbalance

Iron deficiency anaemia is characterized by reduced red blood cells and haemoglobin levels in the blood and small pale cells.

Microcytic anaemia – infants, preschool children, adolescent girls and pregnant women are susceptible.

Hemosiderosis – large deposits of iron are made in the liver, spleen, and mitochondria of cells. May also occur in abnormal destruction of red blood cells as in the case of hemolytic anaemia. It may also occur in prolonged iron therapy.

Haemochromatosis – is an inherited defect in regulation of iron absorption in which high amount of iron are absorbed. Excess iron may be deposited in the lungs, pancreas and heart.

IODINE

Iodine is a non-metallic mineral which humans require in trace amounts for proper development and growth. It exists in most soils, and is taken up by plants which are in turn ingested by humans and animals. Most of the body's stores of iodine are located in the thyroid gland, which requires iodine for the synthesis of thyroid hormones.

Why is iodine necessary?

Iodine is used by the thyroid gland to produce hormones which regulate all key metabolic functions including blood cell production and nerve and muscle function. Because our body heat is primarily derived from muscle metabolism, these hormones also regulate body temperature. Topically applied iodine is also useful to help prevent wounds from becoming infected, and

iodine tablets dissolved in water provide an inexpensive way to purify water in emergency situations.

What are the signs of a deficiency?

Signs of iodine deficiency include an enlarged thyroid gland (goiter), and weight gain, as well as symptoms of **hypothyroidism** such as fatigue and intolerance of cold. Chronic iodine deficiency can lead to health problems such as thyroid gland dysfunction, and neurologic, gastrointestinal, and skin abnormalities. Iodine deficiency in pregnant or nursing mothers can result in deficiency during fetal and child development, and is the most common cause of preventable brain damage in the world.

How much, and what kind, does an adult need?

Adults 18 and older should get 150 mcg daily; pregnant women 18 and older, 220 mcg; and lactating women, 290 mcg.

How much does a child need?

Infants 0-12 months, 50 mcg; Children 1-8 years, 90 mcg; Children /young teens 9-13, 120 mcg; Teens 14-18, 150 mcg.

How do you get enough iodine from foods?

Iodized salt is the main source of iodine in the diet. Plant and animal sea life, such as shellfish, white deep water fish, and brown seaweed kelp, absorb iodine from the water and are great sources of iodine. Garlic, lima beans, sesame seeds, soybeans, spinach are also good sources of iodine.

Are there any risks associated with too much iodine?

Signs and symptoms of acute iodine poisoning include burning of the mouth, throat, and stomach; fever; nausea; vomiting; diarrhea; a weak pulse; and coma. Excessive iodine intake can be toxic to those with kidney disease or tuberculosis, and may increase the risk of thyroid papillary cancer. There is an unhealthy practice of using iodine tablets to "boost" thyroid function, and toxicity can result. You should never take iodine tablets without a physician's recommendation to do so.

ZINC

Zinc is an essential trace mineral, so you get it through the foods you eat. Next to iron, zinc is the most common mineral in the body and is found in every cell. It has been used since ancient times to help heal wounds. Zinc is a mineral that is necessary for the functioning of enzymes and plays a vital role in many biological processes, including the production of cells that help keep the body healthy.

Why is zinc necessary?

1. Zinc helps maintain optimum immune function and boosts immunity, and creates new cells which allow healthy collagen production and wound healing.
2. Zinc also has antioxidant properties', meaning it helps protect cells in the body from damage caused by free radicals. Free radicals may contribute to the aging process, as well as the development of a number of health problems, including heart disease and cancer. Antioxidants can neutralize free radicals and may reduce or even help prevent some of the damage they cause.
3. It is also a component of key enzymes that help preserve vision and protect against age-related vision loss, including macular degeneration.
4. Zinc further plays a role in carbohydrate and protein metabolism, and may be beneficial as a supplement for people with severe diarrhea, sickle cell anemia, gastric ulcers, and acne.
5. In addition, zinc is vital for normal fetal development and the maturation of sperm.

What are the signs of a deficiency?

Signs of zinc deficiency include hair loss, weight loss, delayed wound healing, chronic infection, and rough skin or rashes. Symptoms include poor appetite, depression and mental lethargy.

How much, and what kind, does an adult need?

Adults take 15 mg of zinc daily, or up to 30 mg daily for vegetarians or for those who don't eat many foods of animal origin.

How much does a child need?

Infants 7 months - 3 years should get 3 mg daily; children 4 - 8 years, 5 mg; 9 - 13 years, 8 mg; males 14 years and older 11 mg; females 14 -18, 9 mg; pregnant females 14 -18, 13 mg; and pregnant females 19 and older, 11 mg.

How do you get enough zinc from foods?

Animal foods such as beef, eggs, and oysters (cooked) are a rich source of zinc. The best plant sources are legumes (dried beans, garbanzos, black-eyed peas, lentils, peas, and whole soy products), pumpkin seeds, whole grains and nuts.

Are there any risks associated with too much zinc?

High doses of zinc can actually decrease immunity, interfere with the absorption of copper, leading to copper deficiency, and increase the risk of anemia. Excessive amounts of zinc can also cause nausea, headaches, lethargy, irritability, stomach irritation, and vomiting.

SELENIUM

Selenium is a mineral found in soil, water, and some foods. It is required in trace amounts for normal health, and is an essential element in several metabolic pathways.

Why is selenium necessary?

1. Selenium has antioxidant properties that help the body prevent cellular damage from free radicals, and one of its most valuable roles is as a cofactor of an important antioxidant enzyme in the body called glutathione peroxidase.
2. Selenium also helps support a strong immune system,
3. Regulates thyroid function, and may help reduce the risk of prostate and secondary cancers.
4. It also plays a role in the prevention of cataracts and heart disease.

What are the signs of a deficiency?

A deficiency in selenium can affect thyroid function and lead to diseases such as: Keshan Disease (enlarged heart and poor heart function in children), Kashin-Beck Disease (results in osteoarthropathy), and Myxedematous Endemic Cretinism (results in mental retardation). Symptoms can include muscle weakness and pain.

How much, and what kind, does an adult need?

The Recommended Dietary Allowance (RDA) for adults:

Adult

- 19 and older: 55 mcg
- Pregnant women: 60 mcg
- Breastfeeding women: 70 mcg

How much does a child need?

- Children 1 - 3 years: 20 mcg
- Children 4 - 8 years: 30 mcg
- Children 9 - 13: 40 mcg
- Children 14 - 18: 55 mcg

How do you get enough selenium from foods?

Brewer's yeast and wheat germ, liver, butter, fish (mackerel, tuna, halibut, flounder, herring, smelts) and shellfish (oysters, scallops, and lobster), garlic, whole grains, sunflower seeds, and Brazil nuts are all good sources of selenium. Selenium levels in food depend on how much selenium was in the soil where the food was grown. Selenium is destroyed when foods are refined or processed. Eating a variety of whole, unprocessed foods is the best way to get selenium in your diet.

Are there any risks associated with too much selenium?

High blood levels of selenium can result in selenosis, a condition with side effects including gastrointestinal distress, hair loss, white spots on nails, fatigue, and irritability.

COPPER

Copper is an essential trace mineral, present in all body tissues, which plays a role in the formation of connective tissue, and in the normal functioning of muscles and the immune and nervous systems.

Why is copper necessary?

The human body requires copper for normal growth and health. Copper, along with iron, is a critical component in the formation of red blood cells. Copper also influences the functioning of the heart and arteries, helps prevent bone defects such as osteoporosis and osteoarthritis, and promotes healthy connective tissues (hair, skin, nails, tendons, ligaments and blood vessels).

What are the signs of a deficiency?

Copper deficiency is rare, but can occur in people who are severely undernourished or who have chronic diarrhea. Disorders that impair nutrient absorption, such as Crohn's disease, can also lead to copper deficiency, as can high dietary intakes of iron or zinc. Signs of deficiency include bleeding under the skin, damaged blood vessels, hair loss, pale skin, and an enlarged heart. Symptoms include fatigue and, because copper plays a role in immunity, imbalances can make you more susceptible to infections.

How much, and what kind, does an adult need?

Adult

- For adults 19 years and older: 900 mcg daily
- For pregnant women: 1,000 mcg daily
- For breastfeeding women: 1,300 mcg daily

How much does a child need?

Pediatric

- For infants from birth - 6 months: 200 mcg daily
- For infants 7 - 12 months: 220 mcg daily
- For children 1 - 3 years: 340 mcg daily
- For children 4 - 8 years: 440 mcg daily
- For children 9 - 13 years: 700 mcg daily
- For children 14 - 18 years: 890 mcg daily

How do you get enough copper from foods?

Good food sources include vegetables, legumes, beans, nuts and seeds, mushrooms, shellfish (especially cooked oysters), avocado and whole grains.

Are there any risks associated with too much copper?

Copper is toxic in large amounts, and acute poisoning can lead to nausea, vomiting, diarrhea and even kidney damage, anemia and death. Wilson's disease is a rare (affecting about one in 30,000 people) inherited disorder in which excessive amounts of copper accumulate in the liver or brain. If you have Wilson's disease, or a family history of it, do not take supplements containing copper and get regular testing to monitor your copper levels.

FLUORIDE

Fluorides, commonly known by the collective term fluoride, are compounds containing fluorine, the most reactive element found on the periodic table. The most common fluorides are sodium fluoride and calcium fluoride, which is found naturally in low concentration in bodies of water, especially well water, and in some foods. Fluorides such as hexafluorosilicic acid and sodium hexafluorosilicate are also added to city water supplies to help prevent tooth decay. Sodium fluoride, stannous fluoride, and sodium monofluorophosphate are all fluorine compounds added to toothpaste, also to help prevent tooth decay.

Why is fluoride necessary?

Fluoride can help prevent tooth decay and cavities, and promotes strong teeth and enamel. The addition of fluoride to most public water supplies is credited with a 40 to 60 percent reduction in tooth decay in both children and adults who live in fluoridated communities.

Fluorides also help maintain bone structure. Low doses of fluoride salts may be used to treat conditions that cause faster-than-normal bone loss, such as menopause.

What are the signs of a deficiency?

Tooth decay and poor tooth health are signs of fluoride deficiency, and long-term deficiency can lead to osteoporosis.

How much, and what kind, does one need?

Infants

- 0 - 6 months: 0.01 milligrams per day (mg/day)
- 7 - 12 months: 0.5 mg/day

Children

- 1 - 3 years: 0.7 mg/day
- 4 - 8 years: 1.0 mg/day
- 9 - 13 years: 2.0 mg/day

Adolescents and Adults

- Males ages 14 to 18 years: 3.0 mg/day
- Males over 18 years: 4.0 mg/day
- Females over 14 years: 3.0 mg/day

How do you get enough fluoride from foods?

Food prepared in fluoridated water contains fluoride. Natural sodium fluoride is in the ocean, so most seafood contains fluoride. Tea and gelatin are natural sources of fluoride.

Infants can only get fluoride through drinking infant formulas. Breast milk has a negligible amount of fluoride in it.

Are there any risks associated with too much fluoride?

Consuming excessive fluoride can create chalky white, irregular patches on the surface of teeth, causing the enamel to appear mottled. In very large amounts, fluoride is toxic and can cause gastrointestinal symptoms and sometimes even death. Signs of fluoride overdose include excessive salivation, tremors, weakness, convulsion and a soapy or salty taste in the mouth. In addition, high amounts of fluoride over several years can cause brittle bones, but this is extremely rare